Brief description and motivation

Melting temperature of DNA is the temperature at which half the duplexes are open/close. It is the temperature after which the DNA duplex is unstable and thermally denature.

Several experiments and modeling techniques rely on accurate predictions of this melting temperature for short DNA (<100 base pairs) in the presence of both monovalent (e.g., Sodium) and divalent (e.g., Magnesium) salt in the surrounding buffer. The presence of those positively charged ions (cations) significantly affects the stability of those negatively charged duplexes.

One crucial example of where accurately knowing the melting temperature is important is in Polymerase Chain Reaction (PCR) which simply involves amplifying small amounts of short DNA strands to obtain more samples that can then be used for a more detailed analysis such as in the detection of COVID-19.

Typical PCR buffers involve both monovalent and divalent cations, and optimal amplification is achieved using annealing temperature that is 1-2 degrees Celsius away from the melting temperature of the given duplex (Owc-2008 - see references below).

Hence, it is crucial to predict melting temperatures accurately when designing PCR assays.

Literature Review

Main models and datasets

We rely on an experimental dataset developed in Owc-2008. They combined prior melting experiments of DNA in monovalent dominant buffers with new results involving divalent dominant buffers as well as buffers with both types of cations.

After their own data analysis, they developed a correction formula that predicts melting temperature to great accuracy for Magnesium dominant conditions (Eq. 16 in main text)

They used Linear Regression on a dataset of 680 melting temperature with minimal monovalent salt present to fit 7 parameters. They further corrected some coefficients (Eq. 18-20 in main text) by making them monovalent salt dependent to improve predictions in buffers when both cations are present using a mixed dataset of 456 melting temperatures (this dataset also includes 60 additional reference melting temperatures at monovalent dominant conditions – <u>this is our training/validation set</u>). They tested this model on an independent dataset (Table S2 in supplementary materials) and obtained an average MAE of about 1 degree Celsius (<u>this will be used for testing our models</u>).

For monovalent dominant conditions, they similarly developed another formula in a prior study (Owc-2004), where a separate (divalent-independent) dataset of 580 melting temperatures was used for this regression.

Combining both formulas, they developed a decision tree that can be used (Figure 9 in Owc-2008) to predict melting temperature at any given salt condition using one of their three formulas: 1) Monovalent only formula (Eq. 4 in Owc-2008), 2) Divalent only formula with fixed coefficients (Eq. 16 in Owc-2008), or 3) Divalent only formula with some monovalent-dependent coefficients (Eqs. 16 and 18-20).

However, these formulas are quite complicated as they involve: 1) a decision tree that relies on the ratio concentration ratios, which is not sufficiently motivated, 2) some parameters being salt dependent themselves with complex dependencies, 3) the melting temperature at 1M Sodium that is obtained using the nearest-neighbor parameters [SantaLucia-1998] (yet another model) in general, 4) highly nonlinear dependencies on both [Mono] and [Div] concentrations.

Other proposed models

A new set of experiments, Unz-2010 and Unz-2017, involving force-rupture of DNA hairpins showed quite similar prediction power of melting temperatures when compared to thermal melting results where they use their derived nearest-neighbor parameters to predict the melting temperatures.

Instead of having nonlinear salt dependencies, they suggested a linear-heterogenous method of calculating melting temperatures that resolves some of the aforementioned complications.

Relying on how melting temperature is obtainable from formation enthalpy and entropy the given duplex, they developed a nearest-neighbor set of parameters that have linear (in Log[Salt]) salt dependency **per parameter** with a dinucleotide dependent correction (Table 2 and Eq. 4 in Unz-2017) instead of it being a global salt correction for the entire duplex

However, they were able to predict either monovalent dominant or divalent dominant salt conditions. They also had to resort back to experimental melting temperatures to obtain the individual enthalpies and entropies. A global formula using this approach was not obtainable to the best of my knowledge.

Our goal

Our goal in this project is two-fold:

1. We would like to achieve high prediction accuracy (Low MAE) of melting temperature with mixed salt conditions using a small dataset (516 melting temperature - 12 Main sequences).
2. We *hope* to have a SINGLE global model that is simple/interpretable and captures the most biophysically relevant dependencies of the melting temperature (Low complexity and number of features).

In the EDA notebook, we start by showing prior trends of the melting temperature as a function of sequence (length and composition) as well as salt conditions (monovalent and divalent) to develop (in the Modeling notebook) a 1) more accurate, 2) global, and hopefully 3) easier to interpret (simpler) model of melting temperature of short DNA at ANY given salt condition.

It is indeed a challenge to achieve all, accuracy-generality-interpretability, but let's see how far we can get!

 # Models and Metrics

To handle the first goal, we will be using Mean Absolute Error (MAE) as our metric since it was the main metric used by the prior studies (Owc-2008).

Also, given the limited number of data points and the target accuracy, I think it will be best to use non-linear models that handle such scenarios right away. In particular, we will compare LinearRegression (LR), GeneralizedAdditiveModels (GAM), and DecesionTreeRegressor (DT).

Given the EDA we did in the EDA notebook, non-linear dependencies and salt interactions seem unavoidable, so these models would be our first approach in capturing such complexity and confirm such behavior.

To investigate the second goal, we will test the winning models on multiple variations of the dataset each with a different set of features (Features Engineering) to find *the minimal set* of features needed to generalize to unseen data. This will help us narrow down the most important features and potential salt-feature interactions.

The point of interpretability/simplicity is a challenge, but LR and GAM should make it possible to tackle such challenges.

If these fail, DT would be the next more complex model that captures non-linear effects, but it is also more likely to overfit. We consider it in the comparison for completeness.

 # Features Engineering

We will test different sets of features to select the most important ones.

The best performing model should be the one performing the best in terms of the number of features and the validation-MAE score (Lower=Better, for both).

As explored in the EDA notebook and as seen in literature, the melting temperature is highly dependent on GC-content (fraction of G and C in the duplex), Duplex Length, and Log[Salt], but it will be nice to see how good a model that has other features engineered is doing.

The main motivation of the features used should be its ability to generalize to unseen sequences (beyond those 12 in the training set), so we will try to extract important features of those 12 sequences at hand that can capture their primary properties, yet sharable by similar sequences and able to generalize to different sequences.

In particular, the duplex length will need to be present in all sets of features since it is a *signature property*; however, the GC-content for example can be captured in different ways.

To this end, we compared the models' performance on three sets of features:

1. *The typical set of features - our baseline model*: Similar to the ones usually found in literature, we add the GC-content feature.
2. *The naive set of features*: This will just include the fraction of each nucleotide (A,T,C,G).

3. *The modern set of features*: This will use the fraction of the ten different dinucleotide steps (AA/TT, GC/CG, AC/GT, ...) to capture the suggestion of the more recent papers (Unzip-2010 and Unzip-2017). For simplicity, we will ignore the two terminal contributions (Edge effects) for now (Unzip-2010).

Generally:

- They all include the duplex length, as highlighted before.
- They all include monovalent and divalent salt concentrations in mM.

We will also check if Log-transformation of the salt concentration in M is necessary, following literature trends.

- For each set of features, we will have a Feature*Salt-interaction set.

This will include Monovalent_mM*Feature and Divalent_mM*Feature, for each feature.

	typical_no_salt	naive_no_salt	typical_with_salt	modern_no_salt	naive_with_salt	modern_with_salt
Number of Features	4	6	8	11	14	29
full list of features	[monovalent_mM, Mg2+_mM, length_bp, GC_content]	[f_A, f_G, f_C, length_bp, monovalent_mM, Mg2+_mM]	[monovalent_mM, Mg2+_mM, length_bp, GC_content, monovalent_mM-length_bp, monovalent_mM-GC_content, Mg2+_mM-length_bp, Mg2+_mM-GC_content]	[length_bp, monovalent_mM, Mg2+_mM, f_AA/TT, f_AT/TA, f_CA/GT, f_GT/CA, f_CT/GA, f_GA/CT, f_CG/GC, f_GG/CC]	[f_A, f_G, f_C, length_bp, monovalent_mM, Mg2+_mM, monovalent_mM-f_A, monovalent_mM-f_G, monovalent_mM-f_C, monovalent_mM-length_bp, Mg2+_mM-f_A, Mg2+_mM-f_G, Mg2+_mM-f_C, Mg2+_mM-length_bp]	[length_bp, monovalent_mM, Mg2+_mM, f_AA/TT, f_AT/TA, f_CA/GT, f_GT/CA, f_CT/GA, f_GA/CT, f_CG/GC, f_GG/CC, monovalent_mM-length_bp, monovalent_mM-f_AA/TT, monovalent_mM-f_AT/TA, monovalent_mM-f_CA/GT, monovalent_mM-f_GT/CA, monovalent_mM-f_CT/GA, monovalent_mM-f_GA/CT, monovalent_mM-f_CG/GC, monovalent_mM-f_GG/CC, Mg2+_mM-length_bp, Mg2+_mM-f_AA/TT, Mg2+_mM-f_AT/TA, Mg2+_mM-f_CA/GT, Mg2+_mM-f_GT/CA, Mg2+_mM-f_CT/GA, Mg2+_mM-f_GA/CT, Mg2+_mM-f_CG/GC, Mg2+_mM-f_GG/CC]

Training and Cross-Validation

On top of the different sets of features, we want to know what part of the training dataset is the most important for generalization and whether the whole set of 12 sequences is truly needed or it is possible to leave some sequences or categories out.

To this end, we compared three different methods of cross-validation (CV) using the most important trends found in literature. In particular, we perform GC-content, Duplex Length, and Sequence based leave-out CV.

GC-content CV (LeaveOneCategoryOut_GC CV or LOCO_GC)

The 12 training duplexes come in three main categories of GC content (Low=0.3/Medium=0.5/High=0.7), which gives 3-fold CV:

```
LEAVE-GC-CATEGORY-OUT CV Results:
```

	Linear Regression (Training)	Linear Regression (Validation)	Generalized Additive Model (Training)	Generalized Additive Model (Validation)	Decision Tree (Training)	Decision Tree (Validation)
Typical (4)	2.1609 ± 0.0932	2.3063 ± 0.1954	1.8624 ± 0.0949	2.2275 ± 0.4310	0.0000 ± 0.0000	8.5310 ± 1.3994
Typical with salt interaction (8)	2.1123 ± 0.0903	2.2489 ± 0.2051	1.3256 ± 0.0629	5.5767 ± 4.0776	0.0000 ± 0.0000	8.5124 ± 1.4289
Naive (6)	2.0863 ± 0.0920	4.8847 ± 1.5017	1.4867 ± 0.0728	5.4566 ± 3.4403	0.0000 ± 0.0000	9.5924 ± 2.7709
Naive with salt interaction (14)	2.0387 ± 0.0939	4.8500 ± 1.4948	1.2464 ± 0.0467	5.1128 ± 2.7534	0.0000 ± 0.0000	7.9876 ± 3.3388
Modern (11)	1.7525 ± 0.0968	15.2032 ± 11.9282	1.3290 ± 0.0588	13.7940 ± 13.9846	0.0000 ± 0.0000	8.5421 ± 3.6436
Modern with salt interaction (29)	1.7051 ± 0.0937	15.2354 ± 11.9714	1.2824 ± 0.0024	19.6170 ± 11.0779	0.0000 ± 0.0000	8.6198 ± 3.4537

--

Length CV (LOCO_LENGTH)

The 12 training duplexes come in four main length categories (15/20/25/30bp), which gives 4-fold CV:

LEAVE-LENGTH-CATEGORY-OUT CV Results:

	Linear Regression (Training)	Linear Regression (Validation)	Generalized Additive Model (Training)	Generalized Additive Model (Validation)	Decision Tree (Training)	Decision Tree (Validation)
Typical (4)	2.0285 ± 0.1962	3.6107 ± 1.3127	1.7898 ± 0.0710	3.2125 ± 1.7524	0.0000 ± 0.0000	6.2457 ± 2.3363
Typical with salt interaction (8)	1.9815 ± 0.1981	3.5937 ± 1.3251	1.3341 ± 0.0043	3.6916 ± 1.7731	0.0000 ± 0.0000	6.6640 ± 1.9270
Naive (6)	1.9803 ± 0.1650	4.0086 ± 1.2600	1.4890 ± 0.0027	3.1614 ± 1.6510	0.0000 ± 0.0000	7.7585 ± 4.9361
Naive with salt interaction (14)	1.9335 ± 0.1649	3.9981 ± 1.2699	1.2457 ± 0.0072	3.1161 ± 1.8543	0.0000 ± 0.0000	8.2143 ± 5.3604
Modern (11)	1.7600 ± 0.0185	4.2361 ± 1.1932	1.3419 ± 0.0018	4.1490 ± 1.1544	0.0000 ± 0.0000	5.7101 ± 2.1059
Modern with salt interaction (29)	1.7051 ± 0.0192	4.2231 ± 1.2073	1.3170 ± 0.0096	12.5816 ± 5.0489	0.0000 ± 0.0000	6.8068 ± 2.7963

--

Sequence CV (LeaveOneSequenceOut CV or LOSO)

Probably the easiest of them, this is a 12-fold CV since we have 12 sequences:

LEAVE-ONE-SEQUENCE-OUT CV Results:

	Linear Regression (Training)	Linear Regression (Validation)	Generalized Additive Model (Training)	Generalized Additive Model (Validation)	Decision Tree (Training)	Decision Tree (Validation)
Typical (4)	2.1696 ± 0.0593	2.4955 ± 0.8330	1.8570 ± 0.0266	2.0954 ± 0.5955	0.0000 ± 0.0000	5.9529 ± 1.5767
Typical with salt interaction (8)	2.1252 ± 0.0608	2.4572 ± 0.8477	1.3319 ± 0.0127	1.5152 ± 0.2271	0.0000 ± 0.0000	5.8684 ± 1.5605
Naive (6)	2.1368 ± 0.0581	3.1272 ± 1.0764	1.4785 ± 0.0108	1.9343 ± 0.4815	0.0000 ± 0.0000	8.6417 ± 6.0565
Naive with salt interaction (14)	2.0889 ± 0.0591	3.0947 ± 1.0834	1.2439 ± 0.0090	1.8383 ± 0.6200	0.0000 ± 0.0000	6.6860 ± 5.6568
Modern (11)	1.8131 ± 0.0413	5.6436 ± 2.7061	1.3407 ± 0.0103	5.4559 ± 2.8502	0.0000 ± 0.0000	9.0496 ± 8.0469
Modern with salt interaction (29)	1.7584 ± 0.0438	5.6382 ± 2.7099	1.2250 ± 0.0123	8.7196 ± 3.5731	0.0000 ± 0.0000	8.4411 ± 8.1628

CV discussion and conclusions

Based on the comparison between training/validation MAE results:

There seems to be a train/val gap, especially for DT, which is a clear sign of overfitting using this more complex model.

Similar pattern is seen with more complex features, particularly, the Modern set of features.

<u>Hence, DT and Modern features set are excluded from later comparisons.</u>

Given that the experimental error in melting temperature measurements is around 0.5 degrees and that advanced models give 1-2 degrees MAE (Owc-2008), it feels safe to say that <u>LR and GAM with Typical (with and without salt interactions) and Naïve (without salt interactions) feature* are good preforming (this excludes Naïve features set with salt interactions)</u> and not overfitting.

However, it is hard to tell if this gap is true or just noise for LOCO-GC and LOCO-LENGTH since the number of folds is small and hence the standard deviations of validation MAE are large. <u>*The testing set should (hopefully) provide the final decision on this conclusion.*</u>

Based on the MAE on the validation set:

- GAM is typically outperforming LR, especially with increasing number of folds.

This shows the non-linear nature of the salt dependency.

- The Typical set of features with salt interactions seems to be doing the best overall, so we use it as our best set of features.

This agrees with literature and it is also the least number of features encoding salt interactions, compared to the Naïve one.

- 3. It would have been nice to see the Modern set of features with LR competing with other models, as literature suggested, but the number of features involved in it seems too high given the limited number of data points in our case. *It remains a possibility for future explorations*.
- 4. Log-transformation of salt concentrations did not provide significant improvement (results shown in the Github folder "/Results/Log-transformed). GAM is likely capturing such nonlinearity anyway.

Best (per CV strategy) Model selection:

Given this discussion and that our target is low MAE with low complexity and high interpretability with physical grounding, GAM with the Typical set of features seems the best model for the coming analysis for the three CVs.

It is necessary to use totally unseen data (to the best of my understanding) to decide what type of splitting provides the best generalization.

Hence, we use the lowest validation MAE fitted model from each CV strategy as our model for the final testing:

```
Best LOCO-GC fold: Left-out GC=0.70 with MAE=1.5721
Best LOCO-Length fold: Left-out Length=25.00 with MAE=2.1843
Best LOSO fold: Left-out Sequence=CAGTGGGCTCCTGGGCGTGCTGGTC (L=25, GC-content=0.72) with MAE=1.2313
```

 # Testing

We test the three models on 1) the full training set, 2) a subset of the test set (Table S2 of Owc-2008), and 3) a filtered subset of the test set (in point 2) that include lengths within the training range (15-30bp) to choose the best CV strategy and compare with the complex model developed by Owc-2008. Additionally, we training GAM with the Typical set of features on the full training set (Full GAM) to explore its performance over testing set as well.

On the full training set (516 data points)

Comparison of MAE of our chosen models on the full training data compared to the magnesium correction model:

	Our Model (LOSO)	Our Model (LOCO GC)	Our Model (LOCO Length)	Our Model (Full GAM)	Owc-2008 Correction Model
MAE on Training Data (°C)	1.335808	1.444493	1.548351	1.334051	0.582914

On the test set (39 data points)

Comparison of MAE of our chosen models on the full test data compared to the magnesium correction model:

	Our Model (LOSO)	Our Model (LOCO GC)	Our Model (LOCO Length)	Our Model (Full GAM)	Owc-2008 Correction Model
MAE on full Test Data (°C)	6.428158	6.919956	1.523195	6.591311	0.614954

On filtered test set (20 data points)

Comparison of MAE of our chosen models on the filtered test data compared to the magnesium correction model:

	Our Model (LOSO)	Our Model (LOCO GC)	Our Model (LOCO Length)	Our Model (Full GAM)	Owc-2008 Correction Model
MAE on filtered Test Data (°C)	1.331753	1.332947	1.185532	1.334037	0.378032

<u>The LOCO-LENGTH model outperforms all the other models on different testing sets.</u>

Conclusions and Future directions

It is indeed interesting how the LeaveOneLengthOut is performing the best compared to the other models we trained, even though the CV results were not pointing toward that. One could expect a Voter model that averages the different folds to perform better than the best fold result (the crowd effect), but this will lead to a complex and less interpretable model, so we stick with the model with the best validation MAE.

Compared to the models of Owc-2008, we are not quite there, BUT we are using a SINGLE model instead of a tree of models. Also, our model is performing similar (and sometimes better) compared to other models developed for this purpose (Table 5 of Owc-2008) with the extra advantage of covering all salt types and ranges in one single formula!

The other important aspects of the good performance of our model are how 1) it captures the minimal set of features quite nicely in agreement with literature and 2) it being trained on ONLY about *400 data points*, which is less than half what Owc-2008 used to develop their models (>1000 in total).

Future directions could include:

1. *Using more data:* That will need to be done carefully since we saw how having more data (LOSO and Full GAM) can overfit with the addition of just less than 50 more data points, so one will need to first explore what types of additional data to include without overfitting or hugely increase the data set. One suggestion is having more length-distributed data points rather than multiple sequences of the same length done over multiple salt conditions.
2. *Reexploring the naïve and modern set of features*: It seems to require more data, but if it has the potential of giving us a LinearRegression model that outperforms the standard non-linear ones then it is worth investigating it further (Unzip-2017 and Volo-2018).
3. *Further investigation of splits of the dataset at hand and the best performing model*: It is quite surprising how this model is doing that good with such limited number of data points, so it is important to explore the fitted model smooth functions and what aspects of the additional data lead to overfitting in the other models. *Due to the time limitation, I could not explore this.*

References

1. SantaLucia-1998: J. SantaLucia, A unified view of polymer, dumbbell, and oligonucleotide DNA nearest-neighbor thermodynamics, Proc. Natl. Acad. Sci. U.S.A. 95 (4) 1460-1465, https://doi.org/10.1073/pnas.95.4.1460 (1998).
2. Owc-2004: Effects of Sodium Ions on DNA Duplex Oligomers: Improved Predictions of Melting Temperatures, Richard Owczarzy, Yong You, Bernardo G. Moreira, Jeffrey A. Manthey, Lingyan Huang, Mark A. Behlke, and Joseph A. Walder Biochemistry 2004 43 (12), 3537-3554, DOI: https://doi.org/10.1021/bi034621
3. Owc-2008: Predicting Stability of DNA Duplexes in Solutions Containing Magnesium and Monovalent Cations, Richard Owczarzy, Bernardo G. Moreira, Yong You, Mark A. Behlke, and Joseph A. Walder, Biochemistry 2008 47 (19), 5336-5353, DOI: https://doi.org/10.1021/bi702363u
4. Unz-2010: J.M. Huguet, C.V. Bizarro, N. Forns, S.B. Smith, C. Bustamante, & F. Ritort, Single-molecule derivation of salt dependent base-pair free energies in DNA, Proc. Natl. Acad. Sci. U.S.A. 107 (35) 15431-15436, https://doi.org/10.1073/pnas.1001454107 (2010).
5. Unz-2017: Josep Maria Huguet, Marco Ribezzi-Crivellari, Cristiano Valim Bizarro, Felix Ritort, Derivation of nearest-neighbor DNA parameters in magnesium from single molecule experiments, Nucleic Acids Research, Volume 45, Issue 22, 15 December 2017, Pages 12921–12931, https://doi.org/10.1093/nar/gkx1161
6. Volo-2018: Alexander Vologodskii, Maxim D. Frank-Kamenetskii, DNA melting and energetics of the double helix, Physics of Life Reviews, Volume 25, 2018, Pages 1-21, ISSN 1571-0645, https://doi.org/10.1016/j.plrev.2017.11.012